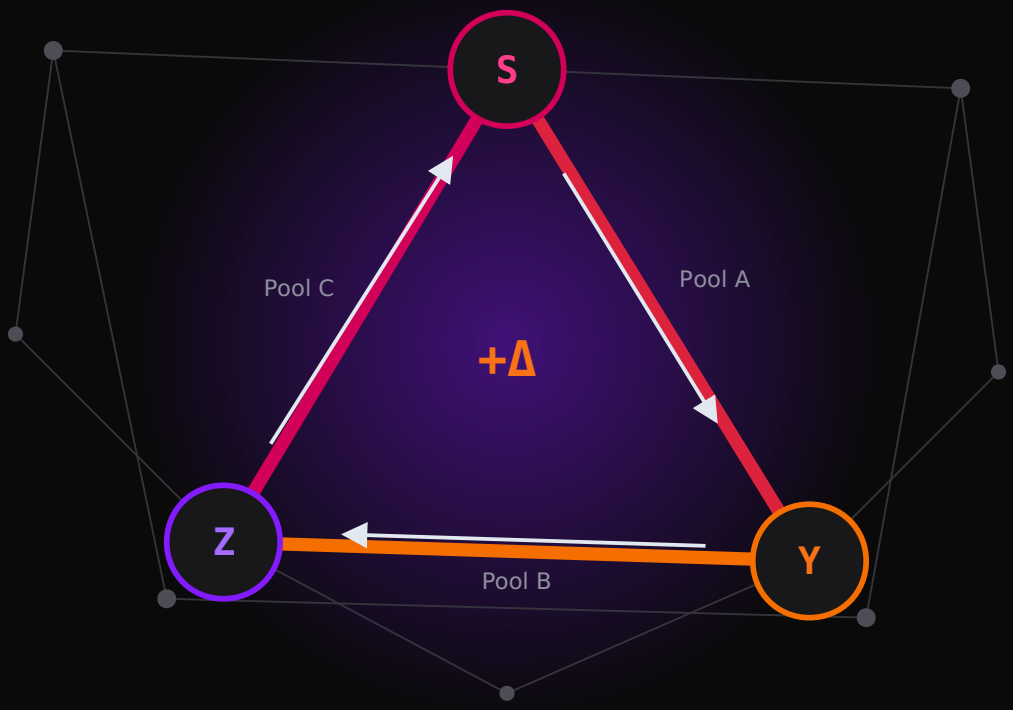
PULSECHAIN · ON-CHAIN ARBITRAGE SCANNER

Arb-Me-Now

Process Manual & White Paper

Methodology, operating procedure, verified parameters and risk model



PulseX v1 · v2 | 9mm v2 | LibertySwap V3 | 2-pool + triangular | BigInt engine | read-only

Version 4.6 · November 2026 · Analysis tool, not financial advice

cryptales.github.io/Arb-Me-Now

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At a glance. Arb-Me-Now reads every pool from one PulseChain block, models each swap with exact integer maths, finds the optimal size for 2-pool and triangular routes, and reports profit before and after estimated gas. It never connects a wallet or sends a transaction.

1. Purpose

Arb-Me-Now finds price differences between decentralised exchanges (DEXs) on PulseChain and tells you whether a round-trip trade would make money **after DEX fees**. It reads live pool data straight from the chain, so all numbers come from a single block. It never signs or sends transactions, and it never asks for a wallet or private key.

2. Architecture

Layer	What it does
Data	Public RPC `rpc.pulsechain.com`. All pool states are read in **one** Multicall3 `tryBlockAndAggregate` call (selector `0x399542e9`), so every price comes from the same block. PulseChain RPC rejects JSON-RPC batch requests, so Multicall3 is required.
Tick data (v4.5)	A second Multicall3 call, pinned to the **same block**, reads `ticks(int24)` (selector `0xf30dba93`) for up to 4 initialised-boundary ticks on each side of every v3 pool, giving `liquidityNet` at each crossing. If this read fails, the engine falls back to single-interval maths.
Discovery	`getPair` / `getPool` lookups on each factory for every pair of tokens in the watch list.
Engine	Pure BigInt maths (`engine.js`) with exact integer swap formulas and no floating-point rounding in the money path. It is unit-tested in Node (`test.js`).
UI	A single self-contained `index.html` hosted on GitHub Pages, with no backend.
External	LibertySwap quote API (bridge checks) and Coinbase spot (ETH/USD reference only).

Venues and verified fees

Venue	Factory	Fee
PulseX v1	`0x1715a3E4A142d8b698131108995174F37aEBA10D`	0.29%
PulseX v2	`0x29eA7545DEf87022BAdc76323F373EA1e707C523`	0.29%
9mm v2	`0x3a0Fa7884dD93f3cd234bBE2A0958Ef04b05E13b`	0.25%
LibertySwap V3	`0x796fcbDC956b85797EFe21145Aa97599B7FB36a6`	0.01 / 0.05 / 0.25 / 1% tiers

All fees were checked on-chain, not taken from documentation.

3. Methodology

3.1 Swap maths

- **v2 (constant product):** `out = in·(10000−fee)·R_out / (R_in·10000 + in·(10000−fee))`, using integer division the same way the pool contract does.

- **v3 (concentrated liquidity, multi-tick since v4.5):** exact swap maths using `sqrtPriceX96` and active liquidity `L`, walking across up to **4 tickSpacing intervals per direction**. When the price crosses a boundary tick, `liquidityNet` is added (moving up) or subtracted (moving down) to get the next interval's `L`. The fee is taken on the gross input. Any trade that would go beyond the last interval read, or into an interval with zero liquidity, is treated as infeasible. If tick data is unavailable, the engine falls back to the conservative single-interval model.

3.2 Safe-depth caps

- A v2 leg may use at most **10% of the input reserve**.
- A v3 leg may not move the price beyond the **4th tick interval** read in its direction (or the current interval in single-interval fallback).

3.3 Route types

1. **Two-pool cycles:** S → Y on pool A, then Y → S on pool B, where A and B are different pools for the same pair.
2. **Triangular 3-hop routes (v4.4):** S → Y → Z → S across three different pools.

3.4 Optimal sizing

For each route, the engine first finds the largest feasible input size by binary search. It then runs a ternary search for the size with the highest profit, which works because profit rises and then falls as size grows. Finally it checks every size in the last narrow window exhaustively. It also evaluates the size you typed in, so you can compare your size with the best one.

3.5 Filtering and pricing

- **Suspicious/dust:** routes where any leg has under the dust threshold in USD depth, or an implausibly large price gap, are hidden by default.
- **USD values:** WPLS/USD comes from the deepest WPLS/stablecoin pool, treating bridged stables as $1. Other tokens are priced through their deepest WPLS pool.
- **Snapshot age:** data is marked stale after 20 seconds.

3.6 Bridge price check (LibertySwap)

`GET apis.libertyswap.finance/v3/swap/quote` gives the real quote from PulseChain to Ethereum (fee 0.5% for USDC and WETH from 369 → 1). The result is compared with the PulseChain purchase price and the Coinbase spot price. **This is not atomic:** the bridge takes minutes and you carry price risk while it runs. The API rate limit is 30 requests per 60 seconds and is enforced on the client.

3.7 Atomic execution: ArbExecutor (v4.6)

`contracts/src/ArbExecutor.sol` (Solidity 0.8.24, Foundry) runs a 2- to 4-leg closed cycle S → … → S in **one transaction**: - **Flash swap on leg 0:** the first pool sends its output first; the remaining legs run inside its callback; leg 0 is repaid in S at the end. No starting capital is needed. - **All-or-nothing:** if the cycle returns less than the amount owed it reverts `NoProfit(returned, required)`; it also reverts if the leftover is below `minProfit`. Only gas can be lost. Profit is sent to the owner. - **Callbacks (verified in on-chain bytecode):** PulseX v1/v2 `uniswapV2Call`, 9mm v2 `pancakeCall`, LibertySwap V3 `libertyV3SwapCallback` (`0x31e2ba55`). - **Safety:** owner-only `execute` / `withdraw`; callbacks accepted

only from the pool currently being swapped while a run is active; deadline; token-continuity check; fee-on-transfer tokens unsupported. - **Leg encoding (uint256):** bits 0–159 pool · 160–167 kind (0 v2, 1 v3) · 168–175 zeroForOne · 176–191 v2 fee bps. Call: `execute(uint256[] legs, uint256 amountIn, uint256 minProfit, uint256 deadline)` (selector `0x29df2c6f`). - **Testing:** 11/11 Foundry tests on a PulseChain fork (live route reverts NoProfit; induced gaps profitable for v2→v2, v3-flash→v2, v2→v3 and a 3-hop; access control, stray callbacks, expiry, open cycles). A JS end-to-end test on an anvil fork drove the scanner's own encoder: contract profit matched the engine's prediction **exactly** for v2→v2 and within **2 wei** with a v3 leg, and a real execute tx paid the owner. Measured `execute` gas (Foundry report): median ≈ 294k, max ≈ 612k.

4. Operating Procedure (step by step)

1. Open https://cryptales.github.io/Arb-Me-Now/.
2. Under **Settings**, choose the start token and the trade size, and turn venues on or off.
   - **Min profit (bps of input):** the smallest return you care about, in basis points of the amount you put in (1 bp = 0.01%; 10 bps = 0.1%; 100 bps = 1%). It is checked against the profit at the **optimal size**, before gas. It only takes effect when **Only rows ≥ min profit** is ticked; otherwise every row is shown. The default is 0, so any profitable route passes. Example: 1,000 WPLS in, 1,002 WPLS out = 2 WPLS profit = 20 bps. The LibertySwap bridge quotes and Triangular 3-hop routes toggles are on by default.
3. Click **Scan**. Check the telemetry bar: RPC is **OK**, the snapshot block is shown, and the snapshot age is under 20 s.
4. Read the results table, which is sorted by profit in USD. Each row shows the route, venues, mid-price gap, best size, profit at the best size, profit at your size, and flags.
5. Click **Inspect** to open the Trade Companion. It shows reserves or v3 state for every leg, the depth caps, and the exact BigInt calculation.
6. **Before acting on any row:**
   - Re-scan to confirm the opportunity is still there in a fresh block.
   - Check the **net after gas** line. Gas is estimated from the live `eth_gasPrice` × 250,000 gas (2-hop) or 350,000 gas (3-hop), valued at WPLS/USD. Real gas use depends on your execution contract.
   - Allow for MEV and front-running risk. A public mempool trade can be sandwiched.
   - Check the pool addresses on scan.pulsechain.com.
7. **Alerts (v4.5):** tick 🔔 **Alert on new profitable route** and allow browser notifications when asked. Pair it with **Auto-refresh every 15 s**. The scanner beeps, shows a banner and sends a notification **once** for each newly profitable route that is ≥ your min bps and still positive after gas (suspicious rows never alert). Alerts only work while the tab stays open.
8. **Execute (v4.6, optional):** deploy your own ArbExecutor once (`forge create src/ArbExecutor.sol:ArbExecutor --rpc-url https://rpc.pulsechain.com --private-key <your key> --broadcast`, run on your own machine). In Trade Companion, enter its address, keep or edit size and min profit (default 50% of expected), click **Build calldata**, then **Dry run** (an `eth_call` from the owner; nothing is signed). If it says "Would succeed", send the calldata from your owner wallet (value 0) before the 120 s deadline.
9. For bridge checks, treat the edge as an upper bound. Gas on both chains, bridge time and slippage reduce it.

5. Interpreting results

Display	Meaning
`not profitable at any size`	Fees are larger than the price gap at every size.
`+x TOKEN` profit	Gross profit at the optimal size, before gas.
`net after gas`	Profit in USD minus the estimated gas cost (green = still profitable).
`v3 multi-tick (≤N intervals)`	v3 legs are sized across up to N tick intervals; liquidity beyond that is not modelled.
`v3 in-interval (conservative cap)`	Fallback when tick data could not be read: size limited to the current tick.
✅ `Would succeed` / ❌ `Would revert NoProfit`	ArbExecutor dry-run result at the latest block.
🔔 banner / notification	A new profitable, net-of-gas route ≥ min bps appeared (deduplicated per route).
`3-hop triangular`	Triangular route. Only profitable ones are listed.
Hidden: suspicious/dust	Thin or anomalous pools have been left out for your safety.

Live findings (1 Nov 2026, block ~27.94M): no profitable two-pool or triangular routes after DEX fees. The best gross gap was 0.22% (WPLS–WETH, PulseX v2 → LibertySwap V3 0.25%), which is below the combined 0.54% in fees. At about 444,000 gwei, the estimated gas is under $0.001 per route, so DEX fees, not gas, are what close the gap. The WETH bridge check showed about +0.17% to +0.30% against Coinbase before gas and slippage. Efficient markets tend to close these gaps fast, so the tool is most useful during volatility.

6. Risks and limitations

- Gas is an estimate (fixed gas units × live gas price). MEV and pending-block state are not modelled.
- v3 depth is limited to 4 tick intervals per direction; deeper liquidity is not modelled.
- ArbExecutor: a public-mempool tx can be front-run or the price can move after the dry run; then it reverts and gas is lost. The contract is unaudited; use small sizes first.
- Alerts run in the browser only: the tab must stay open, and the browser or OS may block notifications or sound.
- The results are a snapshot. By the time a transaction lands, reserves may have changed.
- Bridge round-trips are non-atomic and carry price risk.
- The public RPC can lag or rate-limit you.
- **This is analysis, not financial advice. Always verify on-chain before trading.**

7. Security model

The page is read-only: it uses only `eth_call` and HTTPS GET requests. Execution happens only if you send ArbExecutor calldata from your own wallet. There is no wallet connection, no signing, no stored keys, no tracking and no backend. All code is in one auditable HTML file.

8. Maintenance

- **Source files:** `engine.js` (engine), `ui.js` (UI logic), `body.html` (markup) and `test.js` (Node tests: `node test.js`).
- **Build:** the CSS + `body.html` + `engine.js` (without module exports) + `ui.js` are combined into one `index.html`.
- **Deploy:** commit `index.html` to `main`. GitHub Pages serves it within about a minute.
- **When fees or factories change:** re-verify on-chain and update the constants in `engine.js`.

9. Roadmap

Private/bundled submission (MEV protection) · in-page wallet signing (opt-in) · server-side / Telegram alerts · deeper or adaptive tick ranges · more venues and tokens.

10. Version history

- **v4.6.1:** Fixed stale browser tab title (v4.3 → v4.6); added `arb_build/build.py` build script.
- **v4.6:** ArbExecutor atomic flash-swap contract (PulseX v1/v2, 9mm, LibertySwap V3; 2–4 legs), fork-tested; Trade Companion execution panel with calldata builder and read-only dry run.
- **v4.5:** multi-tick LibertySwap V3 depth (up to 4 intervals per direction, same-block tick reads) and profitability alerts (sound + browser notification, deduplicated).
- **v4.4:** triangular 3-hop routes, gas-adjusted net profit, separate route counts in the summary, and old pages redirected to the scanner.
- **v4.3:** live Multicall3 reserves, verified fees, LibertySwap V3 and bridge quotes, and a BigInt engine.